

Batch: A-4 **Roll No.:** 16010122151

Experiment / Assignment / Tutorial No: 8

TITLE: To design and demonstrate the prototype of the application under consideration.

Objective: To incorporate the UI-UX design principles for the prototype design.

Upon completion of this laboratory exercise, students will be able to:

1. **Analyze** the core functionalities and user flows for the application under consideration.
2. **Apply** UI/UX design principles to create a user-centered prototype for a similar application.
3. **Utilize** prototyping tools to create interactive mockups that simulate key user interactions.
4. **Formulate** design decisions regarding layout, navigation, information hierarchy, and interactive elements.
5. **Evaluate** the usability and effectiveness of their prototype through self-critique and peer feedback.

Expected OUTCOME of Experiment: CO 4

Analyze tools and techniques for prototyping and designing applications

Books/ Journals/ Websites referred:

- **Books:**
 - Goodwin, K. (2009). *Designing for the Digital Age: How to Create Human-Centered Products and Services*. Wiley. (Chapters on user scenarios and storyboarding)
 - Lidwell, W., Holden, K., & Butler, J. (2010). *Universal Principles of Design*. Rockport Publishers. (Look for principles related to storytelling and visualization)
 - Saffer, D. (2010). *Designing for Interaction: Creating Smart Applications and Clever Devices* (2nd ed.). New Riders.
- **Journals:**
 - *Interacting with Computers*
 - *International Journal of Human-Computer Interaction*
 - *Personal and Ubiquitous Computing*
- **Websites:**
 - **UI/UX Design Tools Documentation:**
 - Figma Learn: <https://www.figma.com/learn/>

- Adobe XD Learn & Support:
<https://helpx.adobe.com/xd/get-started.html>
- Sketch Documentation: <https://www.sketch.com/docs/>
- **Design System Guidelines:**
 - Google Material Design: <https://m2.material.io/design> (Relevant for a Google Classroom-like application)
 - Apple Human Interface Guidelines:
<https://developer.apple.com/design/human-interface-guidelines/>
- **UX/UI Best Practices & Patterns:**
 - UX Planet: <https://uxplanet.org/>
 - Nielsen Norman Group (NN/g): <https://www.nngroup.com/> (Search for articles on educational software UX, dashboard design, etc.)

Background Theory:

Prototyping is a crucial step in the UI/UX design process, allowing designers to simulate the functionality and user experience of a product before full development. A prototype can range from low-fidelity (e.g., paper prototypes, basic wireframes) to high-fidelity (e.g., interactive mockups that closely resemble the final product). For this lab, we will focus on creating a mid-to-high fidelity interactive prototype.

Key UI/UX considerations for a the application under consideration:

- **User Roles & Permissions:** How do the interfaces differ for different users including administrators?
- **Information Architecture:** How is content organized (hierarchical flow of information)? Is it intuitive to find specific information?
- **Navigation:** How do users move between different sections of the application? Is the navigation clear and consistent?
- **Core Functionalities:**
 - **Dashboard/Home Screen:** Overview of contents, upcoming events, announcements etc.
 - **Section View:** Specific course content, assignments, classwork, people.
 - **Task Management:** Login/Registration, Search, Add to cart, payment etc.
 - **Communication:** Announcements, comments, private messages, chats.
 - **Resource Sharing:** Uploading and accessing files.
- **Usability Principles:**
 - **Consistency:** Consistent layout, terminology, and interaction patterns across the application.

- **Feedback:** Clear indications of system status and user actions (e.g., prompt messaging and response).
- **Efficiency:** Streamlined workflows for common tasks (e.g., quickly posting an announcement, easily finding an item).
- **Error Prevention:** Design to minimize user errors (e.g., confirmation dialogs for critical actions).
- **Accessibility:** Design considerations for users with diverse needs (e.g., sufficient color contrast, keyboard navigation).
- **Visual Design:** Clear typography, intuitive iconography, appropriate use of color to convey meaning and hierarchy.

This lab will guide you through the process of conceptualizing, designing, and prototyping key screens for a classroom application, applying these principles to create a user-friendly experience.

Laboratory Activities

Duration: 4-5 hours (including research, design, prototyping, and documentation)

Materials:

- Computer with internet access
- Access to a UI/UX design and prototyping tool (e.g., Figma - free tier is sufficient, Adobe XD, Sketch with InVision for prototyping). Figma is highly recommended due to its collaborative features and web-based accessibility.
- Notebook or digital document for planning and reflection.
- Access to an existing look-alike with similar functionality application for inspiration and analysis.

Procedure:

1. Application Scope & User Roles Definition (45 min):

- **Define Core Functionalities:** Identify the 3-5 most critical functionalities your application will support.
- **Define Primary User Roles:** Focus on two main user roles: Customers, Administrators, and Dealers etc.
- **Customer and Administrators:** Briefly outline their primary goals and tasks within the application.
- **Inspiration & Analysis:** Spend some time exploring an existing look-alike application. Take screenshots or make notes on what you like and dislike about its UI/UX, particularly focusing on the core functionalities you identified.

2. Screen Selection & User Flow Mapping (60 min):

- Select **3-4 key screens** that represent the core functionalities and user flows for **both** the Student and Instructor roles. Aim for screens that show distinct interactions.
 - *Example Customer Screens:* Customer Dashboard, Login, Registration, Cart, Payment etc.
 - *Example Administrator Screens:* Admin Dashboard, Add inventory, manage users, manage resources, chat support.
- For each selected screen, briefly map out the primary user flow (e.g., "Customer logs in -> sees dashboard -> clicks on a product -> views details -> add to cart -> make payment etc."). This will guide your prototyping.

2. UI Design & Prototyping (150-180 min):

- Using your chosen UI design tool, create the selected 3-4 screens for **both** the Customer and Administrator roles.
- **Focus on:**
 - **Layout:** Use grids and alignment to create organized and balanced layouts.
 - **Information Hierarchy:** Use typography (size, weight), color, and spacing to guide the user's eye to important information.
 - **Navigation:** Design clear and consistent navigation elements (e.g., sidebars, top navigation, tabs).
 - **Interactive Controls:** Design buttons, input fields, dropdowns, and other controls to be visually distinct, intuitive, and provide clear feedback. Consider different states (e.g., hover, active, disabled).
 - **Color & Typography:** Apply a consistent color palette and typography system across all screens, considering readability and visual appeal.
- **Create Interactive Prototype:** Link your screens together to simulate the user flows you mapped in Step 2. For example, clicking an "Assignment" card on the dashboard should lead to the "Assignment Details" screen.
- **Consider both desktop and mobile views** if time permits, or at least design with responsiveness in mind.

3. Self-Critique and Documentation (45 min):

- **Review your prototype:** Interact with your own prototype as if you were a user.
- **Identify strengths:** What aspects of your design do you think are particularly effective or user-friendly?
- **Identify weaknesses/areas for improvement:** Where might users get confused or encounter friction? What could be done better?

- **Document your design decisions:** For each screen, briefly explain your rationale behind key layout, color, typography, and control design choices, linking them back to UI/UX principles.
- **Export:** Generate a shareable link to your interactive prototype (if using Figma/Adobe XD) or export key screens as images for your lab report.

VLAB :

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Note : use start only once at a time till you reset.

Instructions

1. You will see 6 green colored rectangular blocks on the screen.
2. Your time will start after you press "START" button.
3. Observe all blocks minutely and identify the one that expands in breadth after some time.
4. The moment you observe the difference click on the same block to stop the timer.
5. Record the time and the % noticed difference or the % area difference.
6. Repeat the steps 1-5 and plot the graph between '% area difference' Vs 'number of attempts'.
7. To repeat the same experiment first reset the timer by pressing "RESET" button.



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$A = 34t + 5600$
 $A = 5770$
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---END OF EXPERIMENT---

Note : use start only once at a time till you reset.

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5 s

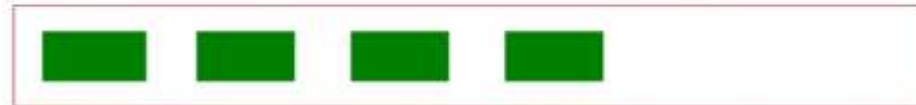
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5 s

A = 34t + 5600
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---END OF EXPERIMENT---

Weber's Law

[Rate Me](#)
[Report a Bug](#)

Note : use start only once at a time till you reach.

Instructions

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7. To repeat the same experiment first reset the timer by pressing "RESET" button.

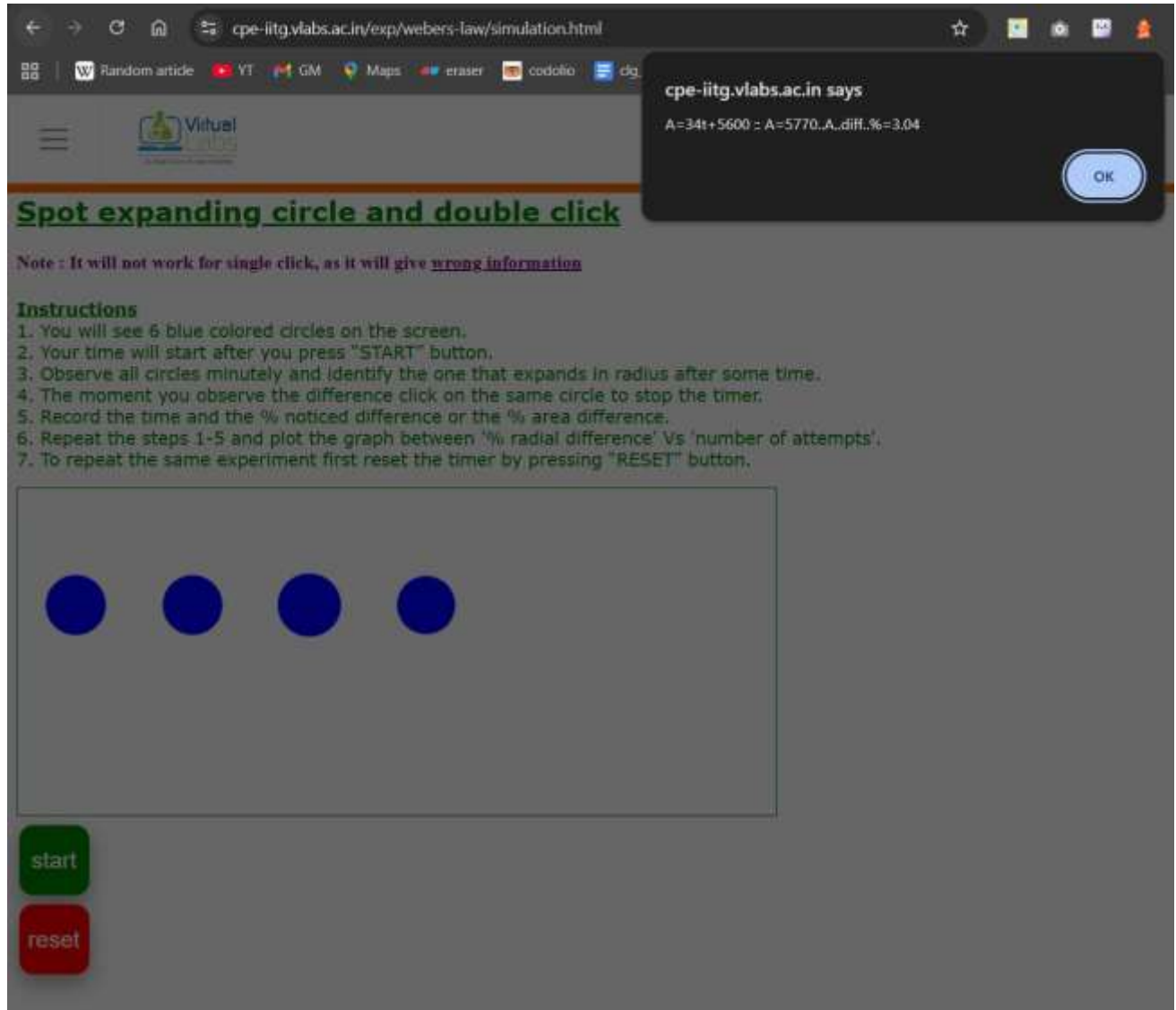


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 Virtual Labs

Spot expanding circle and double click

Note : It will not work for single click, as it will give wrong information

Instructions

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3. Observe all circles minutely and identify the one that expands in radius after some time.
4. The moment you observe the difference click on the same circle to stop the timer.
5. Record the time and the % noticed difference or the % area difference.
6. Repeat the steps 1-5 and plot the graph between '% radial difference' Vs 'number of attempts'.
7. To repeat the same experiment first reset the timer by pressing "RESET" button.



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Spot expanding circle and double click

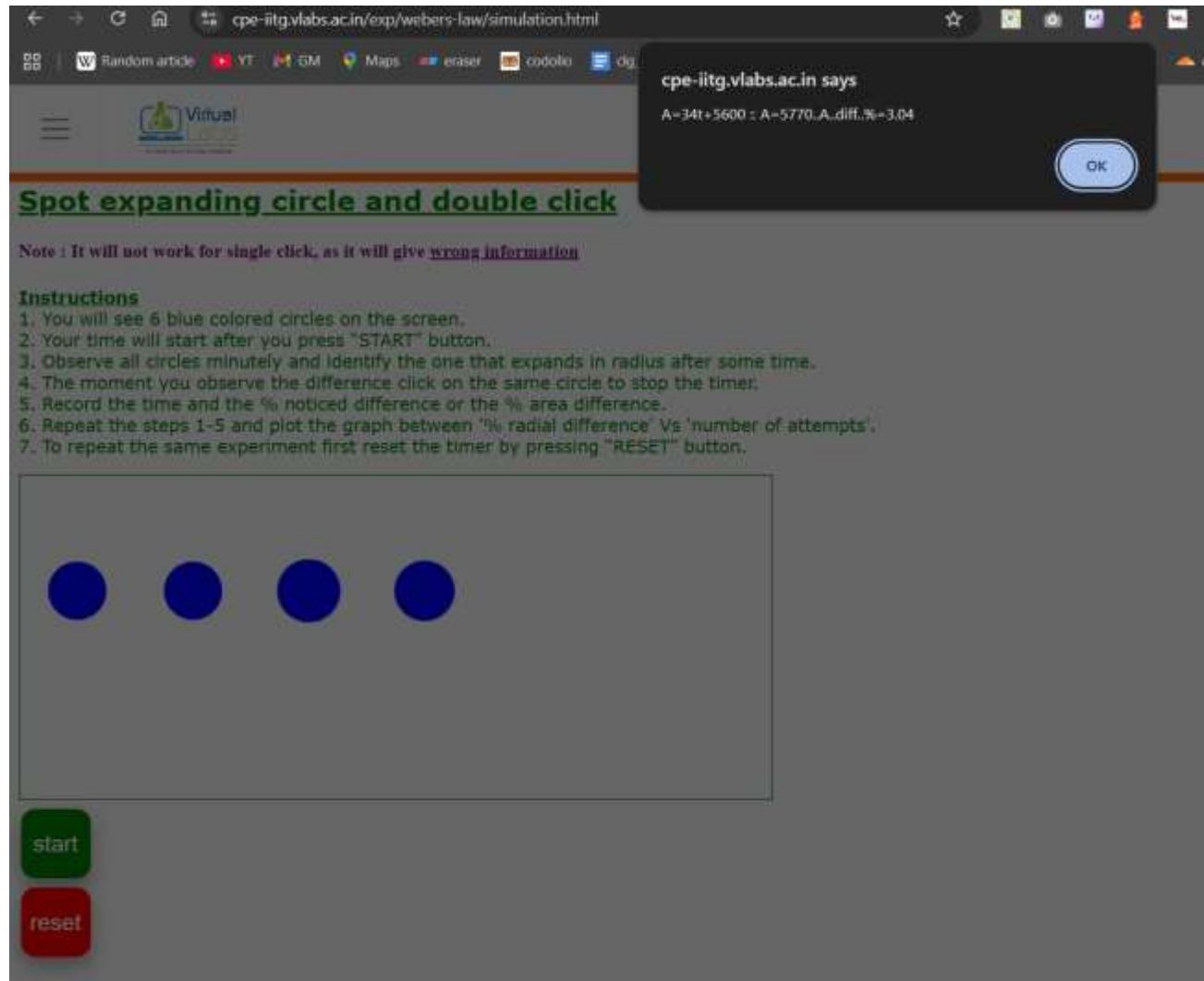
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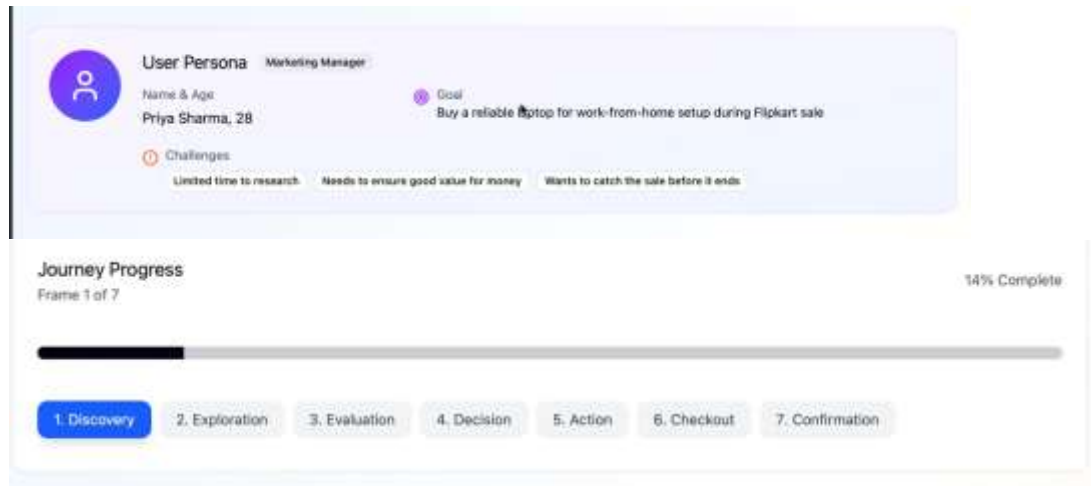
reset



Expt.:

E-commerce Flipkart APP:

User Persona;



User Action

Opens **Flipkart** app after receiving a push notification about a laptop sale

System Response

App displays an attractive sale banner with a countdown timer

User Thinking

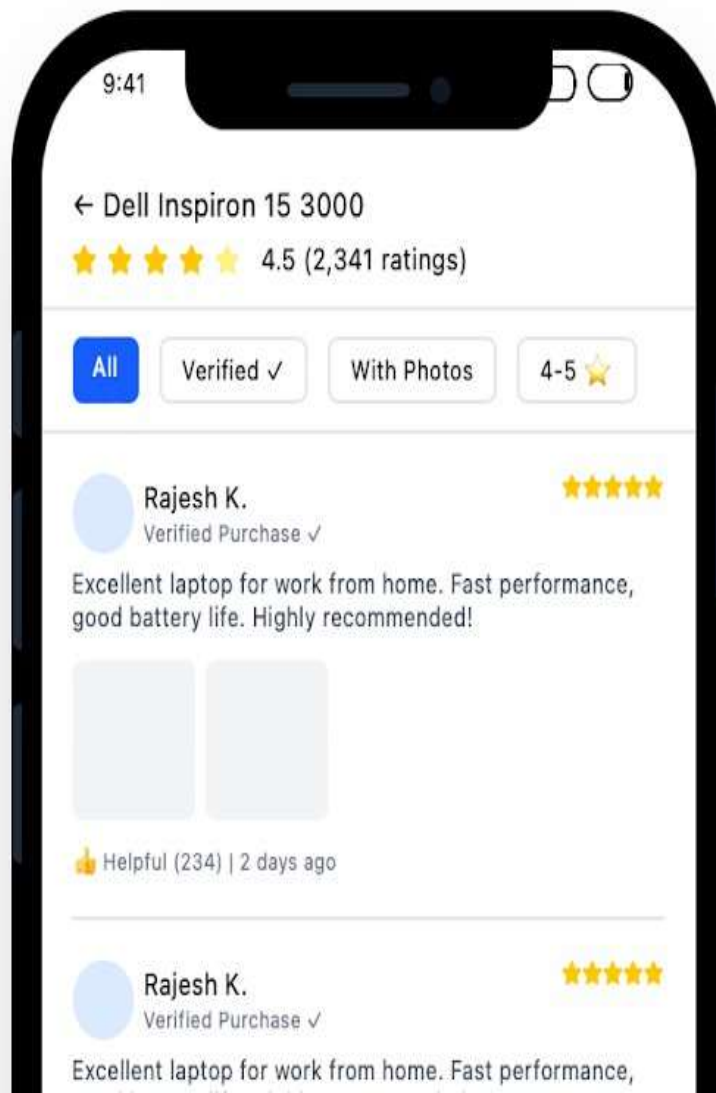
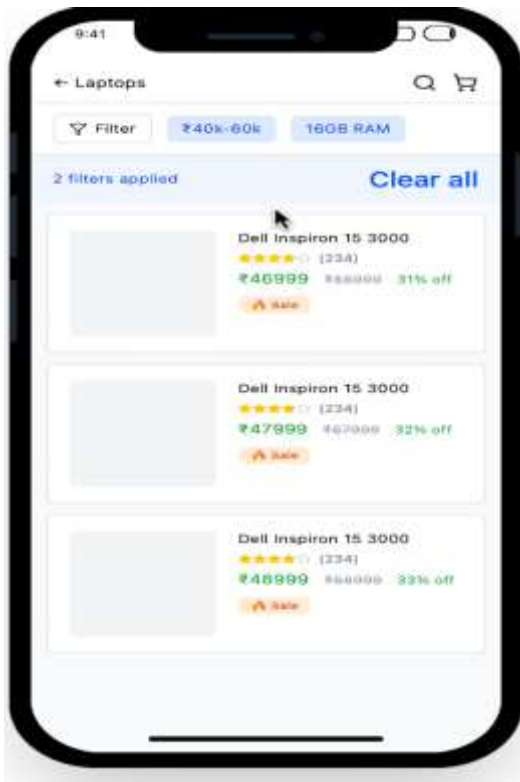
"This could be the perfect time to buy that laptop I've been putting off!"

User Feeling

Excited & Curious

Design Decision

Push notification with urgency creates immediate engagement



User Action

Navigates to '**Laptops**' category and applies filters (Budget: ₹40k–60k, RAM: 16GB)

System Response

Shows filtered results with sale badges, original vs discounted prices, and ratings

User Thinking

"Let me narrow this down... I need something powerful but not too expensive"

User Feeling

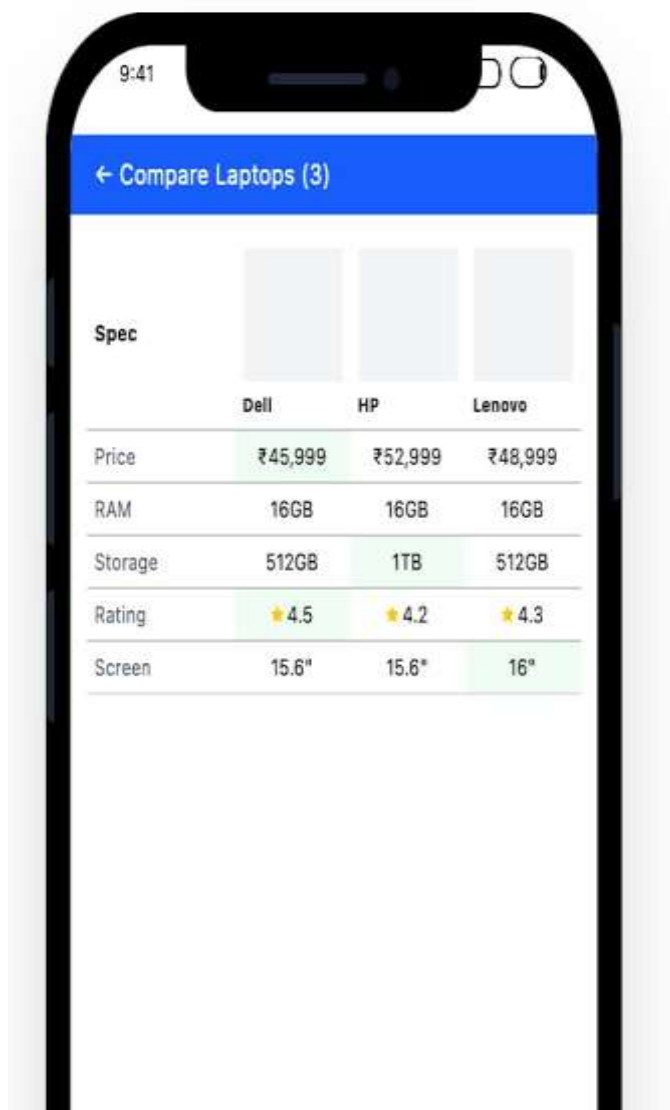
Focused & Engaged

Pain Points

- Too many options can be overwhelming

Design Decision

Clear filters and visual price comparison help decision-making



User Action

Taps 'Compare' button after selecting 3 products, views specifications side-by-side

System Response

Displays comparison table with specs, prices, ratings, and key differences highlighted

User Thinking

"The Dell has better reviews but the HP has more storage... which one should I choose?"

User Feeling

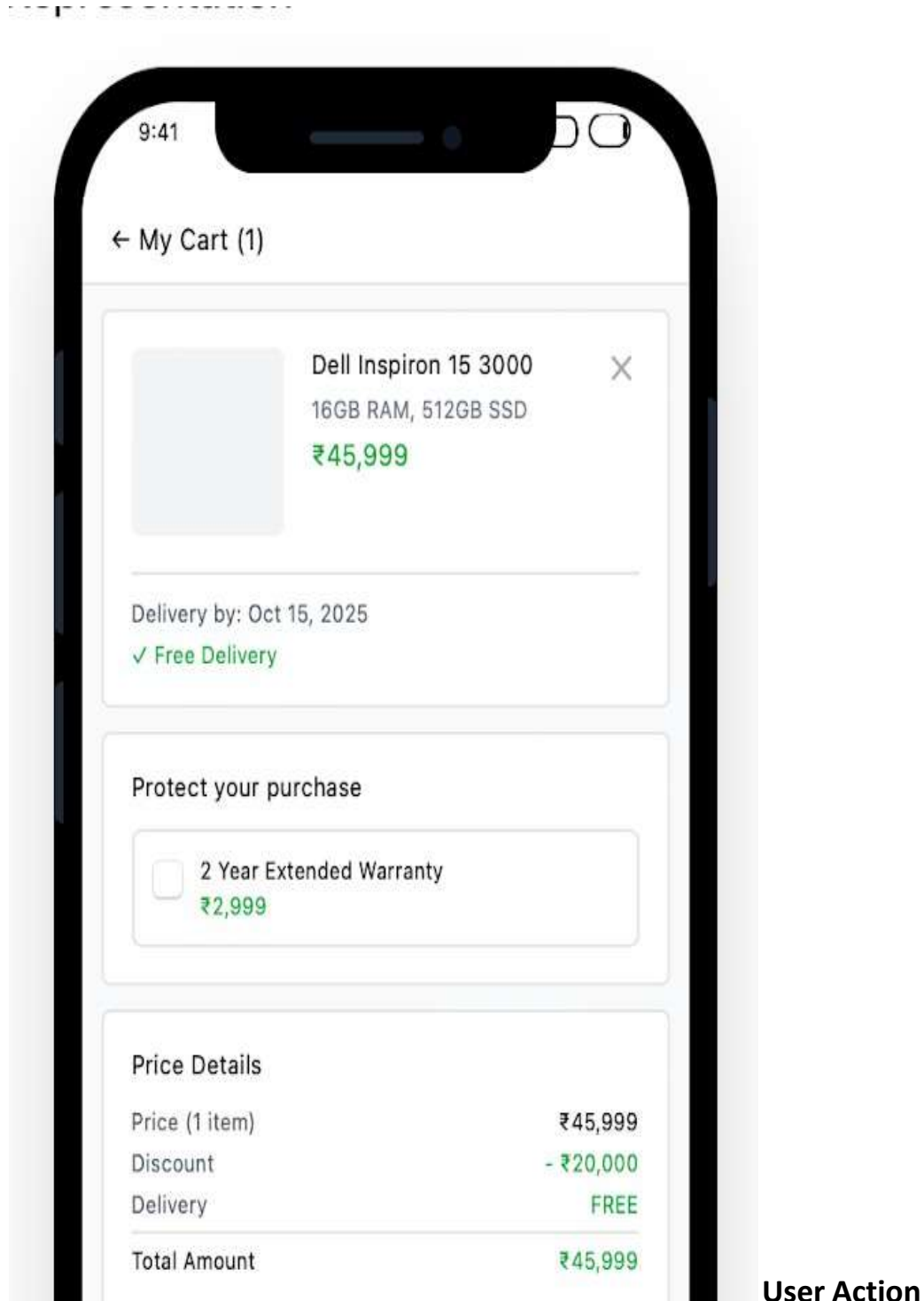
Analytical & Slightly Uncertain

Pain Points

- Decision paralysis with similar products

Design Decision

Comparison feature highlights key differences to aid decision



User Action

Clicks 'Add to Cart' button, then proceeds to checkout

System Response

Cart page shows selected item, offers extended warranty, displays delivery date

User Thinking

"Great! The delivery is in 3 days. Should I get the extended warranty?"

User Feeling

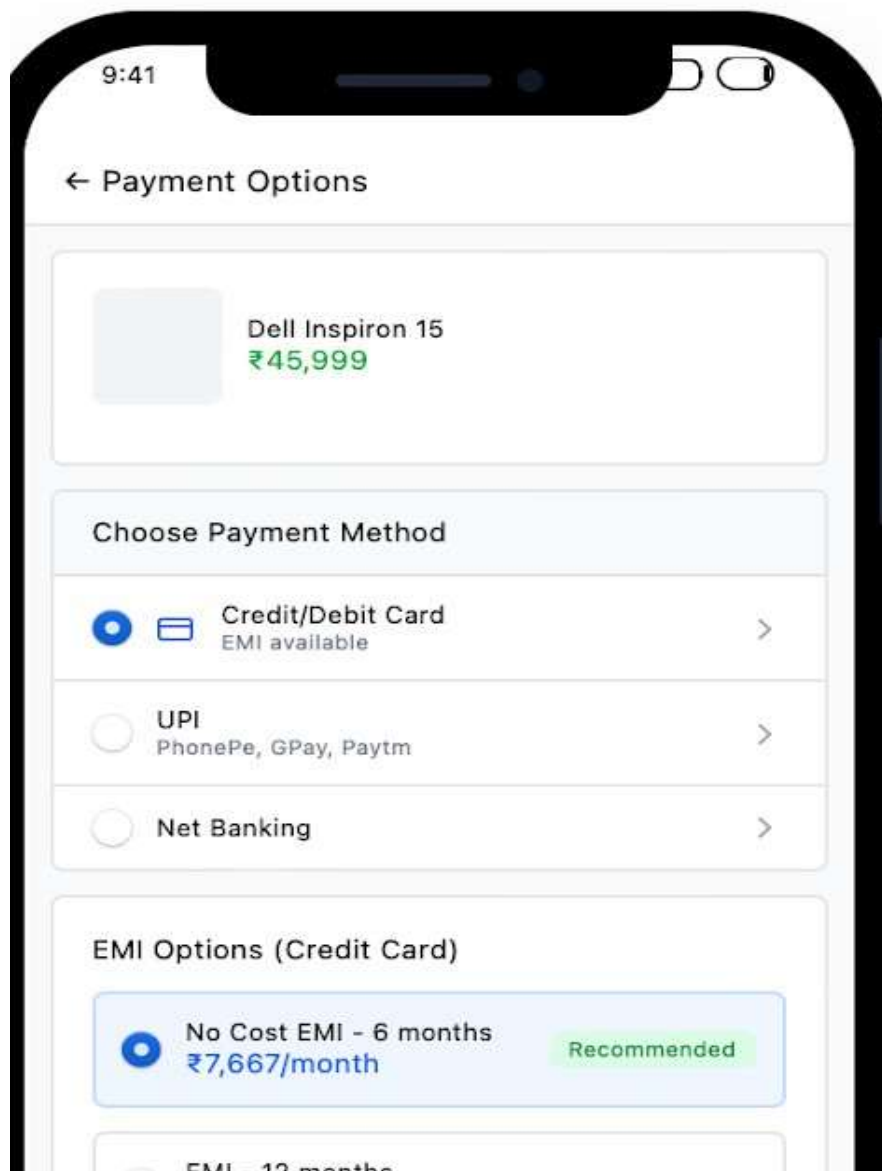
Excited & Slightly Anxious

Pain Points

- Uncertainty about additional services

Design Decision

Clear add-ons with benefits help upselling while maintaining transparency



User Action

Reviews order summary, applies coupon code, selects payment method (Credit Card EMI)

System Response

Shows EMI options with no-cost EMI highlighted, applies discount, updates total

User Thinking

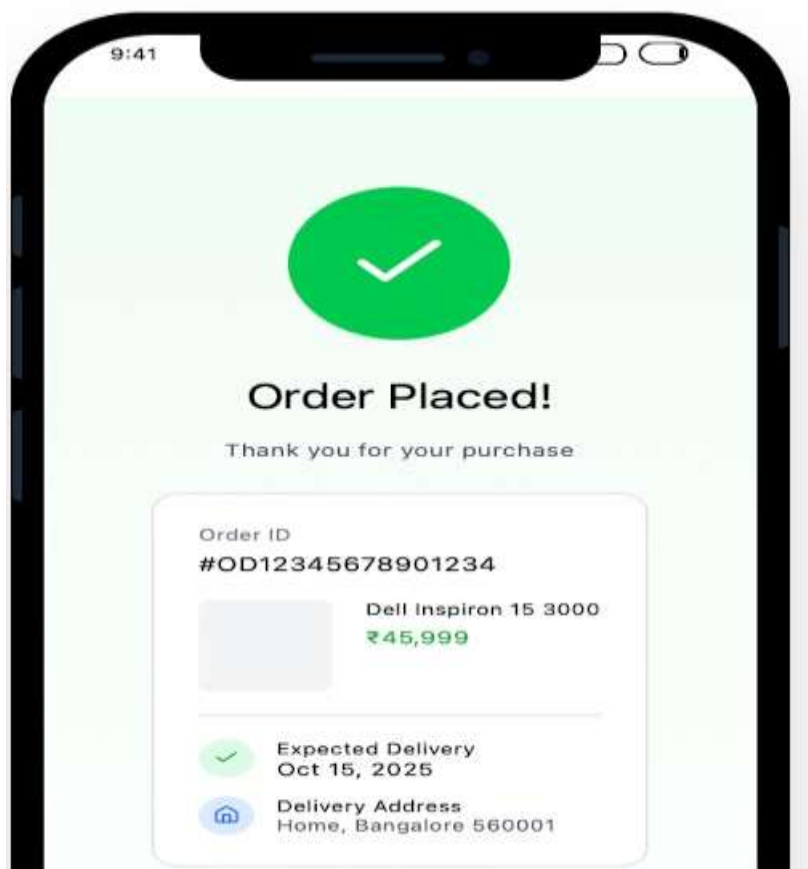
"The no-cost EMI makes this so much easier on my budget!"

User Feeling

Relieved & Happy

Design Decision

No-cost EMI and coupon integration reduce financial pressure



User Action

Completes payment, views order confirmation

System Response

Displays order success animation, order ID, delivery timeline, and tracking option

User Thinking

"Done! I can't wait to start using it. I made a good decision!"

User Feeling

Satisfied & Accomplished

Design Decision

Positive reinforcement with clear next steps ensures user satisfaction

Journey Summary: Major Takeaways

User Experience Highlights

- ✓ Push alerts enhance real-time engagement
 - ✓ Intelligent filters speed up decision-making
 - ✓ Comparison features reduce confusion from too many options
 - ✓ Verified customer feedback establishes trust and reliability
 - ✓ EMI without extra cost eases payment-related worries
-

Pain Points Resolved

- Product overload minimized through guided filtering
 - Indecision reduced via smart comparison tools
 - Trust barriers overcome using verified user reviews
 - Spending anxiety eased with flexible EMI choices
-

Storyboard Evaluation & Reflection

- Is the user journey communicated effectively?
 - Yes, the storyboard flows naturally from discovery to final purchase, showing clear progression and logical connections between each stage.
-

Are emotions represented well?

- Yes, emotions are expressed through icons, descriptive cues, and contextual details that reflect a shift from curiosity to contentment.
-

Is the core value effectively highlighted?

- The storyboard clearly links Flipkart's key features—filters, comparisons, reviews, and EMIs—to specific user challenges, showcasing a seamless end-to-end experience.

Opportunities for Enhancement

- Add more structural visuals within each frame
- Include subtle UI interactions and load transitions

Show alternate outcomes (e.g., user leaving mid-purchase)

- Add time durations to illustrate process efficiency
- Integrate accessibility best practices in design choices

Post-Lab Questions

1. **User Role Differentiation:** Describe how your prototype visually and functionally differentiates between the Student and Instructor interfaces for a common element (e.g., the dashboard or a course stream). Explain the design decisions you made to cater to the specific needs and tasks of each user role.

In the Flipkart prototype, the application visually and functionally differentiates between Customer and Administrator roles to match their specific goals and responsibilities.

Visual Differentiation:

The Customer Interface emphasizes bright blue and white tones with engaging product visuals and banners that promote sales and discovery. In contrast, the Admin Interface adopts muted backgrounds with structured data cards and charts for inventory and order tracking, creating a more analytical environment.

Functional Differentiation:

Customers interact mainly through actions such as browsing products, applying filters, adding items to the cart, and making payments. Administrators, on the other hand, manage products, monitor orders, and handle customer queries through dashboards and CRUD controls.

Design Decisions:

These differences were guided by user-centered design—customers need a fast, visually rich shopping flow, while admins require clarity and control for backend operations. Color contrast, layout density, and feature placement were chosen to minimize confusion and highlight each role's priorities.

2. **Information Hierarchy & Navigation:** Choose one of your designed screens and explain how you used visual design principles (e.g., typography, color, spacing, alignment) to establish a clear information hierarchy. Additionally, describe the navigation elements on this screen and how they guide the user through the application.

Screen chosen: Product Listing Screen (Customer Interface)

Information Hierarchy:

Visual hierarchy was achieved using typography, spacing, and alignment. Product names use medium-weight fonts, while discounted prices are highlighted in bold red for instant visibility. Original prices are shown in a smaller grey font with strikethrough styling to reinforce comparison.

Product ratings and sale badges are visually secondary but grouped closely with product titles to maintain quick readability. Sufficient white space between cards ensures a clean and scannable layout.

Navigation Elements:

A top navigation bar includes the Flipkart logo, a search bar, and category shortcuts for easy access. A breadcrumb trail appears once users enter sub-categories (e.g., Home > Electronics > Laptops). Filters and sorting options are pinned on the left to support focused exploration, while the floating cart icon ensures constant visibility of shopping progress. This layout supports both browsing and decision-making efficiency.

3. **Interactive Control Design:** Select a specific interactive control (e.g., a "Submit Assignment" button, an "Add Announcement" input field) from your prototype. Describe its visual design (color, shape, text, iconography) and how it provides clear feedback to the user upon interaction. How does your design ensure its affordance and discoverability?

Control chosen: "Add to Cart" Button

Visual Design:

The button uses Flipkart's signature yellow color (#FFD814) with bold black text to stand out against the white product background. It has slightly rounded corners (6px radius) to appear approachable, and a cart icon precedes the text for immediate recognition.

Feedback & Interaction:

On hover, the button darkens slightly and shows a subtle lift shadow. When clicked, the button briefly changes to green with a checkmark icon and the text "Added to Cart," providing immediate feedback on the action.

If the user removes an item, the state reverts to the default yellow. This visual change, along with a brief toast message ("Item added to your cart"), ensures the user is always aware of system status.

Affordance & Discoverability:

The button's consistent position at the bottom of each product card, its contrasting color, and universally understood icon afford intuitive interaction. Its visual prominence follows Gestalt and Fitts's Law principles, ensuring it looks and feels like something to click or tap.

4. **Usability Challenges & Solutions:** During your self-critique, identify one significant usability challenge or potential point of confusion you found in your own prototype. Describe the challenge and propose a specific design improvement to address it. Explain how your proposed solution aligns with core UI/UX principles.

Identified Challenge:

During prototype review, it was observed that users occasionally felt overwhelmed by the number of product options on the listing page. The interface lacked immediate guidance for narrowing down results efficiently.

Proposed Solution:

Introduce guided filters and intelligent recommendations that appear contextually (e.g., "Top Laptops under ₹60,000" or "Best-rated options in 16GB RAM"). Filters would dynamically collapse or expand based on user intent, and visual chips at the top of the screen would summarize active filters.

Design Justification:

This modification follows Nielsen's heuristic of reducing cognitive load and improves efficiency of use by allowing users to focus on relevant choices. The guided filtering also enhances the sense of personalization while maintaining the consistency and minimalism principles emphasized in the Material Design Guidelines.